

Associate Architect

9.2 Years Experience

ABOUT ME

Experienced **Mobile and AI Solutions Developer** with strong expertise in **Flutter, Kotlin, and Android Architecture Design**. Skilled in building **scalable, high-performance applications** and **AI-powered platforms** from concept to deployment, with a strong focus on **clean architecture, modular design**, and long-term maintainability. Experienced in integrating **machine learning models, AI voice agents**, and **LLM-driven systems** into production environments. Proficient in developing **end-to-end solutions** using **FastAPI, OpenAI models**, and **real-time communication frameworks**. Passionate about designing efficient architectures that balance **performance, flexibility**, and **user experience** to deliver impactful enterprise and consumer products.

CURRENT COMPANY

2021 - Present

Valueadd Softtech

Responsibilities

- Designed and implemented scalable mobile architectures using Flutter and Kotlin, ensuring modularity, performance, and maintainability.
- AI Interview Agent, AI Voice Agent, and IoT/SHM Monitoring Platforms, aligning technical solutions with business objectives.
- Integrated AI-driven Computer Vision pipelines, LLM-powered automation systems, and smart-glass solutions for real-time field automation, automated interviews, and intelligent customer engagement.
- Built end-to-end IoT data ingestion and analytics systems using MQTT, TimescaleDB, and cloud microservices for large-scale sensor monitoring.
- Led the architecture and development of enterprise applications like Safety App, Home-Works, and Xirify, aligning solutions with business goals.
- Achieved 40% faster performance, 90% fewer App Store rejections, and streamlined white-label client deployments through architectural and process improvements.

SKILLS

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|---|--|---|--|
| • Flutter & Kotlin Mobile App Development | • State Management (BLoC, Cubit, Provider) | • AI & Computer Vision (TensorFlow Lite, MediaPipe, YOLOv8) | • IoT & Sensor Data Processing (MQTT, TimescaleDB, PostgreSQL) |
| • Backend Development (FastAPI, NestJs,) | • LLM & AI Integration (OpenAI APIs, Voice Agents, Interview Agents) | • Architecture Design & Scalable System Planning | • Problem Solving, Innovation, Collaboration, Communication |
| • Effective Communication & Continuous Learning | • Leadership & Mentoring | | |

TOOLS

- TensorFlow Lite, MediaPipe, YOLOv8
 - SQLite, Firebase, Postman, Hive-NoSQL
 - Retell AI
 - Vibe Coding Tools Vercel, Claude, Copilot
- Git, GitLab, GitHub
 - n8n
 - Jira, Confluence
 - Figma , Google Material Design

PROJECTS WORKED ON

1. Golden Data Mortgage Interest Rate Prediction System

Duration of Project: 2 Months
Technology Used: Python, FastAPIs, Retell AI, NextJS, NestJS
Team Size: 3

Project Description:
Golden Data – Voice Agent is an AI-driven solution for the US mortgage domain that predicts home loan interest rates using a machine learning model and automatically contacts customers when their rates are reduced. The system uses a FastAPI-based inference pipeline for predictions and a Retell AI voice agent to place outbound calls. Post-call conversations are analyzed to capture customer interest and intent for follow-up actions.

Role in the Project:
Responsible for integrating the **Retell AI voice agent** with the mortgage interest rate prediction platform to enable automated outbound calling for eligible customers. Implemented the end-to-end workflow connecting the **Context Preparation API, Inference API, and calling services** to ensure accurate and **timely predictions**. Developed backend logic for call triggering based on rate reduction and customer eligibility. Built mechanisms to capture, process, and store post-call conversation outcomes such as customer interest, intent, and follow-up requirements. Ensured smooth data flow between ML services and the voice automation layer through schema validation, error handling, and logging. Supported deployment, testing, and coordination with ML teams to maintain system reliability and performance.

2. IoT Structural Health Monitoring (SHM) System

Duration of Project: 3 Months
Technology Used: NestJS, TimescaleDB/PostgreSQL, AWS (S3/SQS), TypeORM, and TypeScript.
Team Size: 3

Project Description:
Architected a high-performance IoT ecosystem for real-time structural health monitoring, utilizing AWS S3 and SQS to ingest high-frequency telemetry into TimescaleDB. Engineered a Multi-Tiered Continuous Aggregate strategy that downsamples millions of data points into optimized intervals, ensuring sub-second dashboard latency and a 90% reduction in storage via native columnar compression. Delivered a scalable end-to-end pipeline capable of handling 100Hz ingestion with automated lifecycle management and complex analytical reporting.

Role in the Project:

Architected a high-concurrency **time-series database using TimescaleDB** and NestJS, implementing hypertables and native columnar compression to achieve a 90% reduction in storage overhead. Developed an automated data ingestion pipeline using **AWS S3 and SQS** to process high-frequency sensor telemetry asynchronously while ensuring data integrity through custom metadata validation logic. Engineered a hierarchical **continuous aggregate strategy** to enable sub-second dashboard responsiveness for queries spanning millions of records. Designed and optimized complex SQL migration scripts for tiered data downsampling (1min to 6hr) and implemented adaptive API query logic to dynamically fetch the most efficient data resolution. Managed the full development lifecycle of a professional PDF reporting engine, integrating statistical analysis and automated data maintenance policies to ensure long-term system stability and performance.

3. Vision Interview

Agent (AI
Interview
Platform)

Duration of Project: 1.5 Months

Technology Used: NestJS, Next.js, TimescaleDB/PostgreSQL, AWS (S3/SQS), TypeORM, and TypeScript.

Team Size: 3

Project Description:

Vision Interview Agent is an AI-powered recruitment platform that automates resume analysis, conducts real-time video interviews, and generates detailed candidate evaluation reports using Large Language Models. It enables HR teams to manage interviews efficiently through web portals while providing candidates with a seamless AI-driven interview experience.

Role in the Project:

Designed and implemented the end-to-end **AI interview workflow**, integrating resume analysis, interview management, real-time video communication, and automated evaluation. Developed and maintained backend services using FastAPI to handle file uploads, resume-job matching, interview scheduling, and report generation. Integrated **OpenAI GPT models** and Realtime API for intelligent question generation, live interview conversations, and candidate assessment. Built and enhanced React-based HR and candidate portals, including migration of the HR portal from Streamlit to React for better scalability and user experience. Integrated **GetStream Video SDK** and **Vision Agents framework** for seamless real-time interviews. Implemented transcript capture, automated PDF report generation, and secure data storage using Supabase. Ensured system reliability through logging, testing, performance optimization, and deployment support. Collaborated with cross-functional teams to continuously improve platform accuracy, usability, and scalability.

4. Construction

Safety App

Duration of Project: 18-20 Months

Technology Used: Flutter, NodeJs, ReactJs

Tools used: Firebase, Hive-NoSql, Mixpanel

Team Size: 10-12

Project Description:

The Construction Safety App is digital platform designed to streamline on-site safety management through real-time incident reporting, inspection tracking, and compliance monitoring. It enhances communication between field teams and management, ensuring improved accountability and a stronger safety culture across construction projects.

Role in the Project:

Led the end-to-end redesign of the Safety App, focusing on scalable architecture, data model design, and seamless mobile-backend integration. Implemented **Clean Architecture and Repository Pattern** to ensure modularity and maintainability. Designed a **white-label theme structure** enabling client customization within minutes and adopted an **offline-first approach** for reliable performance in low-connectivity environments. Collaborated with backend teams for API design, optimized performance, and improved synchronization logic. Took full ownership of architecture, guided developers on best practices, and ensured a consistent, high-quality user experience across all deployments. Contributed to defining **core modules and reusable components** to accelerate multi-client rollouts and reduce maintenance overhead. Coordinated closely with QA and product teams to validate feature behavior, resolve defects, and maintain release quality.

4. HomeWorks

Duration of Project: 6 Months

Technology Used: Flutter, NestJs, ReactJs

Tools used: Firebase, Mixpanel, MonoRepo

Team Size: 10

Project Description:

HomeWorks is a home services marketplace that connects homeowners with verified service providers. It enables users to manage property details, book and track home maintenance services, and allows agents and market centers to facilitate bookings and earn commissions.

Role in the Project:

In the HomeWorks project, I was responsible for designing and implementing a **monorepo architecture** that **supports three different mobile applications** — Owner, Business Manager, and Agent. I worked on developing **reusable UI components, shared utilities, and common business logic** to ensure consistency and **reduce development time** across all apps. My role also involved collaborating with the backend team for API integrations, managing state efficiently, and implementing role-based access controls for different user roles. I focused on maintaining **clean architecture**, optimizing performance, resolving bugs, and ensuring smooth synchronization between apps. Additionally, I coordinated with QA and business teams for testing and release cycles while guiding team members on best practices for **shared module development**.

5. Xirify

Duration of Project: 9 Months

Technology Used: Flutter, NodeJs, ReactJs

Tools used: Firebase, Mixpanel, Ux-Cam

Team Size: 8-12

Project Description:

A unified platform that brings groceries, home services, and local essentials under one roof. Simplifies daily needs by connecting users with nearby vendors and service providers through a single app. Enhances convenience and engagement for both customers and local businesses.

Role in the Project:

Contributed to the **product re-engineering** of Xirify, a hyperlocal marketplace app that connects users with nearby stores and service providers for daily essentials and on-demand needs. Took ownership of improving the app's usability and performance through the creation of **reusable widgets** and **common UI components**, ensuring design consistency across multiple modules. Collaborated with the product and design teams to enhance the overall **user experience and visual appeal**. Integrated **Mixpanel** and **UXCam** to capture detailed **user behavior analytics** and derive actionable insights for product improvement. Leveraged analytical data to drive iterative UI enhancements that significantly improved user engagement and conversion rates. Focused on optimizing app navigation and interaction flows to simplify the ordering process. Played a key role in the **UI revamp**, resulting in a modern, clean interface that boosted **app downloads to thousands** and increased daily order volume with **total orders crossing 50K** in a short time span. Ensured scalability and flexibility by adopting **component-based architecture** for faster updates and maintainability. Successfully delivered a **Progressive Web App (PWA) version using the same codebase**, achieving seamless cross-platform accessibility. Maintained close collaboration with QA and backend teams to ensure smooth releases, quality performance, and a unified user experience across all platforms.

OTHER PROJECTS

Worked on multiple products across AI, social networking, and real estate management domains as a Senior Software Engineer. Contributed extensively to a native Android project focused on **Computer Vision** and **Smart Glass development**, where I implemented **real-time on-device inference using TensorFlow Lite**, deployed optimized CV models (emotion detection, PPE detection, pose/hand tracking), and built **low-latency local device video streaming pipelines** using WebRTC, Wi-Fi Direct, and custom TCP-based H.264 streaming. Enabled hands-free smart-glass workflows with real-time detection overlays, frame-by-frame analysis, and optimized performance for edge devices.

Additionally, contributed to **Beacon**, a UK-based location-driven social and dating application, and **Client Care**, a digital platform that manages construction site activities and customer interactions from pre- to post-possession. Delivered high-quality, scalable mobile applications with a strong emphasis on usability, performance, and intelligent automation.

WORK HISTORY

SSBA Innovations	1.2 Years		June 2020 - Aug 2021		Software Engineer
Mohar Technologies	2 Year		May 2018 - May 2020		Jr. Software Engineer
Credosync Infosoft	0.9 Years		Mar 2017 - Jan 2018		Associate Mobile Developer
Manomayy Solutions	0.9 Years		Sept 2015 - May 2016		Associate Software Engineer

APP LINKS

[Play Store](#)
[App Store](#)